Q5:Comparative Evaluation of RAG Architectures: Summary

1. Introduction

The study compares four RAG architectures (Sparse/BM25, Dense, Hybrid, and Fusion Retrieval) against a zero-shot baseline in which the model answers without retrieving documents. It focuses on finance and e-commerce, where numerical precision, information freshness, and regulatory auditability matter most.

2. Experimental Design and Evaluation Metrics

The evaluation uses 500 questions, 75% answerable and 25% deliberately unanswerable, to test whether systems recognize insufficient evidence instead of generating unsupported answers. Six dimensions are measured:

- Faithfulness: whether claims are supported by retrieved documents.
- Answer Relevance: whether the response addresses the question.
- Context Precision: whether retrieved passages are relevant.
- Context Recall: whether the relevant information is found.
- Auditability: whether claims trace to valid citations.
- Latency: time required to generate a response.

GPT-4o (cloud-hosted) and Llama-3-8B-Instruct (on-premises) were used, with generator settings controlled to isolate the effect of the retrieval architecture.

3. Comparison of Four RAG Architectures

Sparse Retrieval (BM25) ranks documents by lexical matching using term frequency and inverse document frequency. It is fast, transparent, and explainable, and suits exact-match needs such as legal clauses and regulatory terms. It struggles with synonyms, paraphrasing, and multilingual queries.

Dense Retrieval embeds queries and documents as vectors and retrieves by semantic similarity, finding relevant passages even when vocabulary differs. It offers better recall and multilingual potential, but needs more computation and its similarity scores are harder to interpret and audit.

Hybrid Retrieval runs BM25 and dense retrieval in parallel, merges results with Reciprocal Rank Fusion (RRF), and can rerank passages. It balances lexical precision with semantic coverage and auditability, at the cost of greater engineering complexity, index maintenance, and latency.

Fusion Retrieval expands the question into several sub-queries, gathers evidence through multiple paths, and may use Fusion-in-Decoder (FiD) so the generator attends to many passages. It achieves the strongest faithfulness and answer relevance, but with higher latency, compute demands, context-window pressure, and complexity.

4. Experimental Results

Retrieval grounding substantially improves performance over zero-shot generation.

Architecture	Faithfulness	Answer Relevance	Context Precision	Context Recall	Auditability	Latency (ms)
Zero-shot	0.58	0.61	0.42	0.38	0.35	680
Sparse	0.70	0.76	0.66	0.62	0.71	880
Dense	0.74	0.80	0.63	0.78	0.73	1,020
Hybrid	0.80	0.83	0.69	0.75	0.79	1,110

Architecture	Faithfulness	Answer Relevance	Context Precision	Context Recall	Auditability	Latency (ms)
Fusion	0.82	0.84	0.71	0.77	0.81	1,230

- Sparse is the fastest RAG architecture (880 ms) and retrieves exact-match passages well (context precision 0.66).
- Dense has the highest context recall (0.78), suiting semantically varied queries.
- Hybrid performs strongly and evenly across all dimensions.
- Fusion leads in faithfulness (0.82), answer relevance (0.84), and auditability (0.81), but has the highest latency (1,230 ms).

5. Deployment Considerations

Choosing an architecture means balancing quality, speed, and cost.

- **Sparse:** best for efficiency, exact terminology, and transparent matching.
- **Dense:** best for paraphrases, semantic variation, and multilingual queries.
- **Hybrid:** the recommended general-purpose choice.
- **Fusion:** best when coverage and reliability outweigh speed and cost, as in high-stakes financial research and regulatory systems.

Cloud deployments offer managed infrastructure and usage-based costs; on-premises deployments give more control over sensitive data but require infrastructure investment. Maintaining document timestamps, source URLs, and regularly updated indexes is also important.

6. Limitations and Future Research

The study focuses on finance and e-commerce, so results may not generalize to healthcare, education, or legal technology. It relies heavily on automated RAGAS metrics, which cannot fully replace human assessment. Results may vary with different language models, embedding models, index configurations, and context-window sizes, and cost and environmental impact were not assessed.

Future work should explore dynamic indexing and adaptive pruning, retrieval-aware decoding, joint retriever-generator training, long-context models with stronger claim-level citation alignment, cross-domain and human-in-the-loop evaluation, multi-objective optimization of quality, latency, and cost, and safety audits for bias, leakage, and compliance.

7. Conclusion

All four architectures improve factual performance over zero-shot generation, each with distinct trade-offs: sparse favors transparency and exact matching, dense improves semantic coverage, hybrid is the most balanced general-purpose option, and fusion maximizes factual fidelity at the expense of latency and complexity.

No single architecture is optimal for every application. The choice should depend on accuracy requirements, query characteristics, auditability obligations, latency constraints, and available resources.